



SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

CLASS

CHEMISTRY
JC2 Preliminary Examination
Paper 4 Practical

9729/04
25 Aug 2017
2 hr 30 min

Candidates answer on the Question Paper

Additional Materials:

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.
Give details of the practical shift and laboratory in the boxes provided.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough work.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.
Qualitative Analysis Notes are printed on pages

Shift
Laboratory

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part questions.

For Examiner's Use	
1	
2	
3	
4	
TOTAL	/ 55

- 1** You are to determine the concentration of aqueous copper(II) sulfate by titration. The concentration of Cu^{2+} ions in a solution can be found by reaction with an excess of aqueous iodide ions to produce iodine. The amount of iodine formed can be found by titration with thiosulfate ions, $\text{S}_2\text{O}_3^{2-}$.

FB 1 is aqueous copper(II) sulfate, CuSO_4 .

FB 2 is $0.100 \text{ mol dm}^{-3}$ sodium thiosulfate, $\text{Na}_2\text{S}_2\text{O}_3$.

FB 3 is aqueous potassium iodide, KI .

starch indicator

(a) Method

- (i)**
1. Fill the burette with **FB 2**.
 2. Using a pipette, transfer 25.0 cm^3 of **FB 1** into a conical flask.
 3. Use an appropriate measuring cylinder, transfer 10 cm^3 of **FB 3** to the same conical flask.
 4. Titrate this mixture with **FB 2** until the colour of the mixture changes from brown to yellow-brown. An off-white precipitate will also be present in the flask throughout the titration.
 5. Add 5 drops of starch indicator.
 6. Continue the titration until the blue-black colour of the starch-iodine complex just disappears leaving the off-white precipitate.
 7. Record your titration results in the space provided below. Make certain that your recorded results show the precision of your working.
 8. Repeat points **1** to **7** as necessary until consistent results are obtained.

Results

Initial burette reading / cm^3	0.00	0.00
Final burette reading / cm^3	25.10	25.10
Volume of FB 2 used / cm^3	25.10	25.10

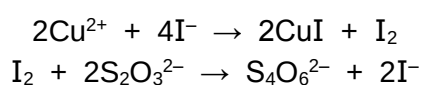
[5]

- (ii)** From your titrations, obtain a suitable volume of **FB 2** to be used in your calculations. Show clearly how you obtained this volume.

$$\text{Volume of FB 2} = \frac{25.10 + 25.10}{2} = 25.10 \text{ cm}^3$$

volume of **FB 2** =[1]

- (b) (i)** The equations for the formation of iodine and its reaction with thiosulfate ions are given below.

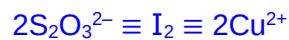


Calculate the amount of thiosulfate ions, $\text{S}_2\text{O}_3^{2-}$, present in the volume of **FB 2** in **(a)(ii)**.

$$\text{amount of S}_2\text{O}_3^{2-} = \frac{0.1 \times (a)(ii)}{1000} = \frac{0.1 \times 25.10}{1000} = 2.51 \times 10^{-3} \text{ mol}$$

$$\text{amount of S}_2\text{O}_3^{2-} = \dots\dots\dots[1]$$

- (ii) Using the equations above, calculate the amount of Cu^{2+} ions present in 25.0 cm^3 of **FB 1**.



$$\text{amount of Cu}^{2+} = (b)(i) = 2.51 \times 10^{-3} \text{ mol}$$

$$\text{amount of Cu}^{2+} = \dots\dots\dots[1]$$

- (iii) Calculate the concentration, in mol dm^{-3} , of copper(II) sulfate in **FB 1**.

$$\begin{aligned} \text{concentration of CuSO}_4 &= \frac{(b)(ii) \times 1000}{25.0} = \frac{2.51 \times 10^{-3} \times 1000}{25.0} \\ &= 0.100 \text{ mol dm}^{-3} \end{aligned}$$

$$\text{concentration of CuSO}_4 = \dots\dots\dots[1]$$

- (c) Two students repeated the experiment but each obtained different values for the concentration of CuSO_4 .

The students each suggested possible improvements.

Student 1 suggested that a larger volume of potassium iodide, **FB 3**, should be added.
Student 2 suggested that the contents of the conical flask should be filtered before titration.

Comment on the effectiveness of **each** of these possible improvements.
Explain your answers.

Student 1 No effect because KI (FB 3) already in excess.

Student 2 Not effective because some iodine stays on the filter paper.

[2]

- (d) The maximum error in each burette reading is $\pm 0.05 \text{ cm}^3$.
The maximum error in each titration is therefore $\pm 0.10 \text{ cm}^3$.
The 25.0 cm^3 pipette used in this titration, is labelled with an error of $\pm 0.06 \text{ cm}^3$.

Explain why the maximum error when 25.0 cm^3 of solution is run from a pipette is only $\pm 0.06 \text{ cm}^3$ and not $\pm 0.12 \text{ cm}^3$.

.....
The mark on the pipette was only read once.
.....

[1]

- (e) Considering the maximum errors from pipette and burette, calculate the total percentage error in one titration.

$$\text{error from pipette} = \frac{0.06}{25.0} \times 100\% = 0.24\%$$

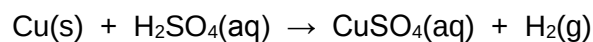
$$\% \text{ error from burette} = \frac{2 \times 0.05}{25.70 (\text{volume from 1 titration})} \times 100\% = 0.39\%$$

$$\text{Total \% error} = 0.24 + 0.39 = 0.63\%$$

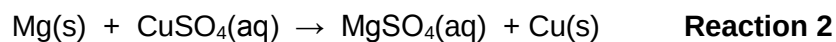
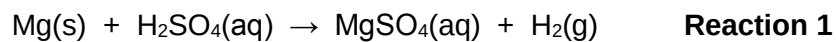
[2]

[Total: 14]

- 2 You are to determine the enthalpy change of reaction, ΔH , for the reaction shown below.



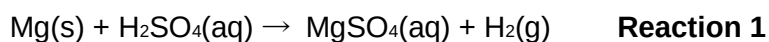
Since copper is an unreactive metal it does not react directly with dilute acids. You will therefore need to find the enthalpy change of reaction for two reactions that do occur. The equations for these two reactions are below.



You will carry out experiments to find the enthalpy changes for each of **Reaction 1** and **Reaction 2** and use these values to calculate the enthalpy change for the reaction of copper with sulfuric acid.

TURN OVER FOR EXPERIMENTAL METHOD

Determining the enthalpy change for Reaction 1



I (a) Method

FB 4 is 1.00 mol dm⁻³ sulfuric acid, H₂SO₄.

FB 5 is magnesium powder, Mg.

Read through the method before you start any practical work and prepare a suitable table for your results.

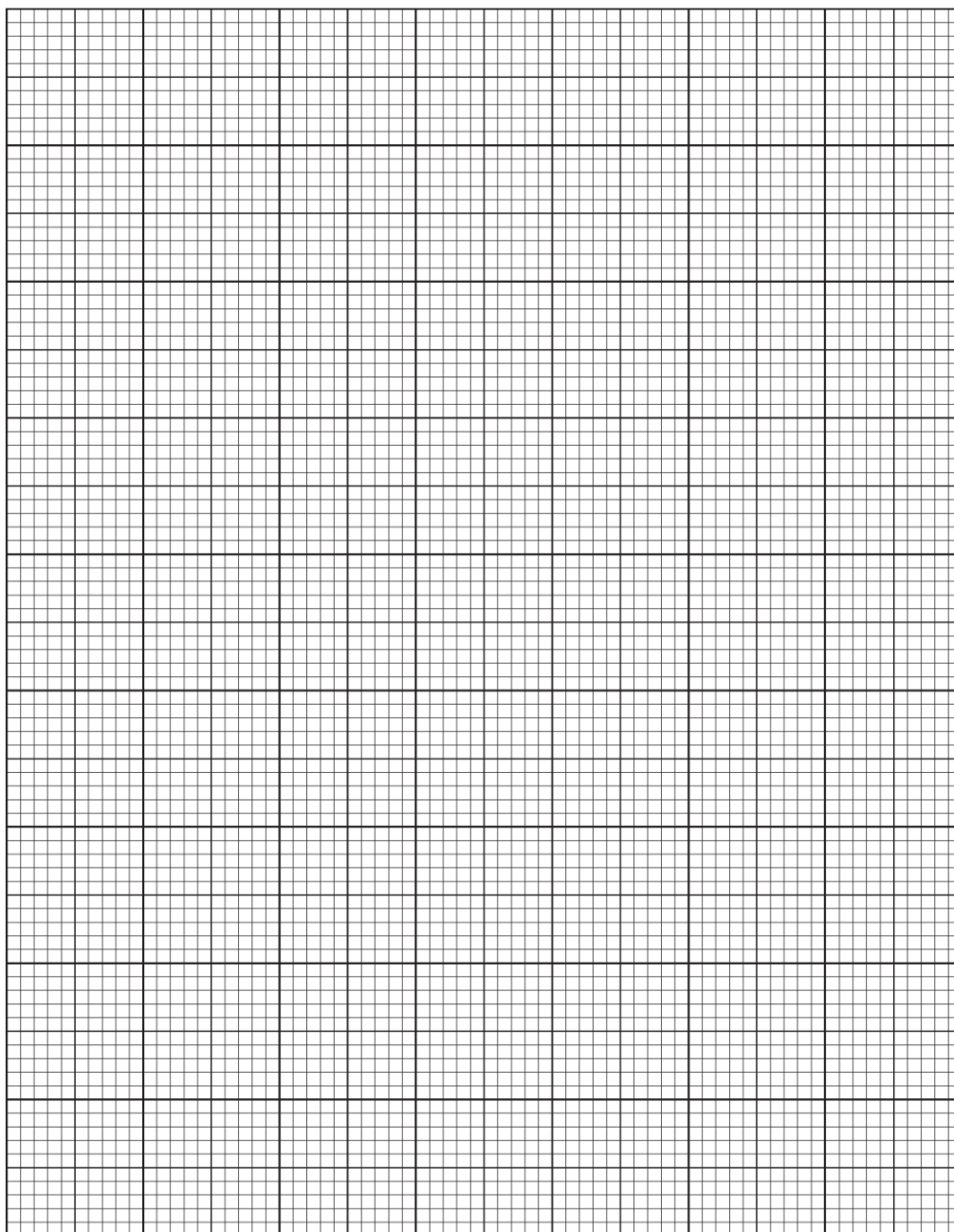
1. Weigh the stoppered weighing bottle containing **FB 5**. Record the mass.
2. Support the styrofoam cup in the 250cm³ beaker.
3. Use the measuring cylinder to transfer 25cm³ of **FB 4** into the styrofoam cup.
4. Measure the temperature of **FB 4** in the styrofoam cup and start the stop watch. Record this temperature as being the temperature at time = 0.
5. Measure, and record, the temperature of this **FB 4** every half minute for 2 minutes.
6. At time = 2.5 minutes, add **FB 5** to the acid and stir carefully to reduce acid spray.
**Caution : Inhalation of the fumes may cause choking.*
7. Measure the temperature of the mixture in the cup at time = 3 minutes and then every half minute up to time = 7 minutes.
8. Continue stirring occasionally throughout this time.
9. Weigh the weighing bottle that had contained **FB 5**. Record the mass.
10. Calculate and record the mass of **FB 5** added to the sulfuric acid.
11. Discard the used styrofoam cup.

Mass of weighing bottle and FB 5 before transfer /g	
Mass of weighing bottle and residual FB 5 /g	
Mass of FB 5 used /g	0.195

Time / min	Temperature / °C
0	
0.5	
1.0	
1.5	
2.0	
2.5	
3.0	
3.5	
4.0	
4.5	
5.0	
5.5	
6.0	
6.5	
7.0	

(b) (i) On the grid below plot a graph of temperature (y-axis) against time (x-axis).

[2]



(ii) Complete the graph by inserting **two, straight** lines of best fit:

- one to show the temperature up to time = 2.5 minutes,
- one to show the temperature after time = 2.5 minutes.

[1]

(iii) From your graph, use the two straight lines of best fit to calculate the change in temperature at time = 2.5 minutes.

temperature change =[1]

(c) Calculations

- (i) In the reaction in **(a)**, the sulfuric acid was in excess. Without carrying out any additional tests, what observation could you have made during your experiment to confirm this?

.....
All the magnesium / solid dissolved / disappeared or all solid / Mg has
gone / been used up or no solid / Mg left
.....

[1]

- (ii) Calculate the energy change that occurred during the reaction in **(a)**.

[Assume that 4.2 J is needed to raise the temperature of 1.0cm³ of solution by 1.0°C.]

Energy change = $25 \times 4.2 \times b(\text{iii}) = \text{XXX J}$

energy change =[1]

- (iii) Use your answer to **(ii)** to calculate the enthalpy change, in kJ mol⁻¹, for the reaction between sulfuric acid and magnesium.

[Ar: Mg, 24.3]



$$\Delta H = - \frac{(ii) \times 24.3}{1000 \times \text{mass of Mg}} = - \text{xxx kJ mol}^{-1}$$

enthalpy change for **Reaction 1** =[1]

Determining the enthalpy change for Reaction 2



(d) Method

FB 6 is 1.00 mol dm⁻³ copper(II) sulfate, CuSO₄.

FB 7 is magnesium powder, Mg.

Read through the method before you start any practical work and prepare a suitable table for your results.

1. Weigh the weighing bottle containing **FB 7**. Record the mass.
2. Support the styrofoam cup in the 250cm³ beaker.
3. Use the measuring cylinder to transfer 25cm³ of **FB 6** into the styrofoam cup.
4. Measure the temperature of **FB 6** in the styrofoam cup and record the temperature.
5. Add the **FB 7** to the **FB 6** in the cup and stir the mixture constantly.
6. Measure and record the maximum temperature reached during the reaction.

7. Calculate and record the maximum temperature change that occurred during the reaction.
8. Weigh the weighing bottle that had contained **FB 7**. Record the mass.
9. Calculate and record the mass of **FB 7** added to the copper(II) sulfate.
10. Empty the contents of the styrofoam cup into the bottle labelled **waste**.

Mass of weighing bottle and FB 7 before transfer /g	
Mass of weighing bottle and residual FB 7 /g	
Mass of FB 7 used /g	

Initial temperature / °C	
Maximum temperature / °C	
Maximum temperature change /°C	

[2]

(e) Calculations

- (i) Show, using a suitable calculation, that the copper(II) sulfate was in excess in the reaction.

[A_r : Mg, 24.3]

$$\text{Amount of CuSO}_4 = \frac{25 \times 1}{1000} = 0.025 \text{ mol}$$

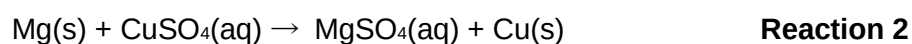
$$\text{Amount of Mg} = \frac{\text{mass of FB 7}}{24.3} = \text{xxx mol}$$

Hence CuSO₄ is in excess.

[1]

- (ii) Hence, calculate the enthalpy change, in kJ mol⁻¹, for the reaction between magnesium and copper(II) sulfate.

[Assume that 4.2 J is needed to raise the temperature of 1.0cm³ of solution by 1.0°C.]

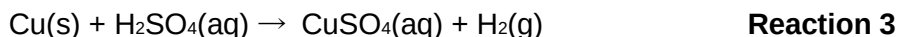


$$\Delta H = \frac{25 \times 4.2 \times \Delta T}{\text{amt of Mg from (i)}} = \text{xxx kJ mol}^{-1}$$

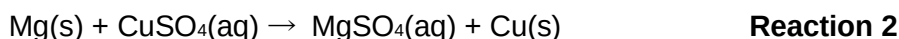
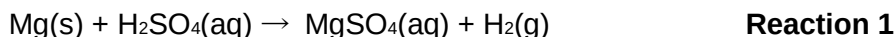
enthalpy change for **Reaction 2** =[1]

Enthalpy change for Reaction 3

Reaction 3 is shown below.

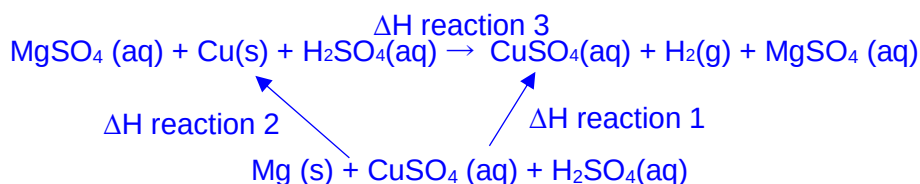


- (f) Use your values for the enthalpy changes for **Reactions 1** and **2** to calculate the enthalpy change for **Reaction 3**.



Show clearly how you obtained your answer.

(If you were unable to calculate the enthalpy changes for **Reactions 1** and **2**, you should assume that the value for **Reaction 1** is -444 kJ mol^{-1} and that the value for **Reaction 2** is -504 kJ mol^{-1} . Note: these are not the correct values.)



$$\Delta H \text{ reaction 3} = \Delta H \text{ reaction 1} - \Delta H \text{ reaction 2} = \text{xxx}$$

enthalpy change for **Reaction 3** =[1]

- (g) (i) The method you used to determine the enthalpy change for **Reaction 1** was more accurate than the method you used to determine the enthalpy change for **Reaction 2**. Suggest why the method used for **Reaction 2** was less accurate. Explain your answer.

- Lower ΔH and so higher % error
- No correction made for loss of heat on cooling
- Some bubbles / gas / H_2 in reaction 2 so wrong reaction taking place
- Not all Mg reacts / reaction does not go to completion in 2 (so not all energy released)
- Reaction 2 slower so more heat loss

[1]

- (ii) A student suggested that the accuracy of the method used for **Reaction 2** could be improved by using a larger volume of copper(II) sulfate. Is this a correct suggestion? Give a reason for your answer.

- No; since (larger volume of solution means) smaller ΔT OR
- Yes; since there would be a smaller T rise so less heat would be lost.

[1]

[Total : 19]

3 Qualitative Analysis

At each stage of any test, you are to record details of the following.

- Colour changes seen
- The formation of any precipitate
- The solubility of such precipitates in an excess of the reagent added

Where gases are released, they should be identified by a test, described in the appropriate place in your observations.

You should indicate clearly at what stage in a test a change occurs.

No additional tests for ions present should be attempted.

FB 8 is a solution containing 2 cations and 1 anion.

(a)

Test	Procedure	Observation
1	To 1 cm depth of FB 8 in a test tube, add aqueous ammonia until it is in excess.	• (grey) Green ppt formed, insoluble in excess NH_3. • Green ppt turned brown on contact with air
2	To 1 cm depth of FB 8 in a boiling tube , add aqueous sodium hydroxide until it is in excess. Then heat the boiling tube gently with care. Keep the mixture for Test 3 .	• Green ppt formed, insoluble in excess NaOH. • Gas evolved that turned moist red litmus paper blue. • Green ppt turned brown (grey) upon heating.
3	To the mixture from Test 2 , add 1 cm depth of aqueous hydrogen peroxide. Shake the boiling tube gently.	• Ppt / solid turned brown / red-brown. • Effervescence observed. Gas evolved relights a glowing splint / glows brighter.
4	To 1 cm depth of FB 8 in a test tube, add aqueous barium chloride, followed by dilute hydrochloric acid.	• White ppt formed, insoluble in HCl.

Using your observations above, identify the ions in **FB 8**. Give evidence to support your conclusion.

Fe^{2+} : From Tests 1 and 2, green ppt formed, turned brown on contact with air, insoluble in excess NaOH and NH_3

(b) NH_4^+ : From Test 2, gas evolved that turned moist red litmus paper blue when warmed with NaOH.

SO_4^{2-} : From Test 4, white ppt formed when BaCl_2 is added, insoluble in HCl.

[3]

(c) What type of reaction has taken place in **Test 3**? Explain your answer.

Redox reaction.

Explanation:

- Iron(II) ions converted / oxidised to iron(III) ions
- Hydrogen peroxide converted / reduced to oxygen gas.
- Hydrogen peroxide decomposed to give oxygen gas

[1]

(d) **FB 9** is an aqueous solution, containing one cation and one anion. Carry out the following tests in test-tubes. Complete the table by recording your observations.

Test	Procedure	Observations
5	To 1 cm depth of FB 9 , add aqueous sodium hydroxide until it is in excess.	• No ppt formed.
6	To 1 cm depth of FB 9 , add a piece of magnesium ribbon.	• Effervescence observed. • Gas evolved extinguished lighted splint with a 'pop' sound.
7	To 1 cm depth of FB 9 in a test tube, add 2 drops of aqueous silver nitrate, followed by aqueous ammonia until in excess.	• White ppt formed, soluble in NH_3 (aq) to form a colourless solution.

[2]

(e) Using your observations above, identify the ions in **FB 9**. Give evidence to support your conclusion.

H^+ : hydrogen gas evolved, extinguished lighted splint with 'pop' sound.

Cl^- : White ppt formed, soluble in NH_3 (aq) to form a colourless solution.

[2]

[Total: 12]

- 4 Many transition metal complex ions are coloured. It is possible to use this property to determine the concentration of a solution of a coloured ion. A few cm³ of the solution is placed inside a *spectrometer*.

A *spectrometer* measures the amount of light that is absorbed when a specific wavelength of visible light is shone through a coloured solution. It does this by comparing the amount of light passing through the sample with the amount of light passing through the pure solvent. The amount of light absorbed is expressed as an absorbance value. The more concentrated the solution, the higher the absorbance value.

Beer-Lambert's Law states that the absorbance values, A , is directly proportional to the concentration of absorbing species, c , as shown below. The general Beer-Lambert's Law is usually written as,

$$A = \epsilon cl$$

Where ϵ is the molar extinction coefficient and l is the path length, which is usually 1.0cm.

This equation can be used to calculate the absorbance value when the concentration of copper(II) ions is known.

You may assume that you are provided with the following in the subsequent parts of the question.

- **FB1** from Question 1
- solid hydrated copper(II) sulfate, $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$;
- access to a spectrometer and instructions for its use;
- graph paper;
- the apparatus and chemicals normally found in a school or college laboratory.

- (a) The spectrometer is set to use the wavelength of light that is absorbed most strongly by the copper(II) ions.

Suggest a colour in the visible spectrum from which a suitable wavelength of light might be chosen. Explain your answer.

Orange. Cu^{2+} is blue in colour. Hence, it will absorb colours that are complementary to **blue**.

[1]

- (b) Spectrometry can be used to determine the concentration of a solution of aqueous copper(II) sulfate. A series of known, but different, concentrations of copper(II) sulfate is prepared. A spectrometer is used to measure the absorbance of each solution.

According to Beer-Lambert's Law, a graph of absorbance against concentration is then plotted. This graph is known as *calibration* line.

The experiment is then repeated using a solution of unknown concentration. By comparing the absorbance of this solution with the calibration line, the concentration of copper(II) ions in the unknown solution can be determined.

Propose a simple plan on how you would prepare

- a 100.0cm³ of 1.00 mol dm⁻³ standard aqueous copper(II) sulfate;
- a suitable range of diluted solutions of accurate concentrations

You are to show detailed calculations and suitable tables (where appropriate) in your answer.

- (i) a 100.0cm³ of 1.00 mol dm⁻³ standard aqueous copper(II) sulfate;

Preparation of the 100 cm³ of 1.00 mol dm⁻³ aqueous copper (II) sulfate standard solution

Amount of CuSO₄·5H₂O needed = (100/1000) × 1 = 0.100

Mass of CuSO₄·5H₂O needed = 0.100 × (63.4+32.1+16×4+5×18) = 24.95 g

1. Weigh accurately about **24.95 g** of CuSO₄·5H₂O using a clean, dry weighing bottle using a mass balance by recording the mass of the CuSO₄·5H₂O and weighing bottle as shown in the table below.

Mass reading table

Mass of weighing bottle with CuSO ₄ ·5H ₂ O / g	<i>x</i>
Mass of empty weighing bottle / g	<i>y</i>
Mass of CuSO ₄ ·5H ₂ O / g	<i>x - y</i>

2. Transfer the weighed solid into a 100 cm³ beaker and dissolve it using distilled water and stir.
3. Transfer the solution and washings into a **100 cm³ volumetric flask**.
4. Reweigh the empty weighing bottle and record its mass.
5. Make up to the mark with distilled water and shake well to obtain a homogeneous solution.

- (ii) a suitable range of diluted solutions of accurate concentrations, keeping the total volume of each solution to be constant at 20.00cm³

Preparation of a suitable range of diluted standard solutions

6 different solutions with different concentrations of CuSO₄ (aq) were prepared, as shown in the following table:

Table 2

Solution	Volume of 1.0 mol dm ⁻³ of CuSO ₄ (aq) / cm ³	Volume of deionized water / cm ³	Total volume / cm ³	[CuSO ₄] / mol dm ⁻³	Absorbance value
1	20.00	0.00	20.00	1.000	
2	10.00	10.00	20.00	0.500	
3	5.00	15.00	20.00	0.250	
4	2.50	17.50	20.00	0.125	
5	1.50	18.50	20.00	0.075	
6	0.50	19.50	20.00	0.025	

1. To prepare 0.500 mol dm⁻³ of CuSO₄ (aq), add 10.00 cm³ of 1.00 mol dm⁻³ CuSO₄ standard solution into a 50 cm³ beaker using a burette.
2. Using a separate burette, add 10.00 cm³ of deionised water into the beaker and stir using a glass rod to obtain a homogenous solution.
3. Repeat the above steps to obtain diluted solutions of different concentrations using the volumes in Table 2.

[5]

- (c) Using the solutions prepared in (b), the absorbance value of each copper(II) sulfate solution can be determined and hence a calibration line can be obtained via the spectrometer.

Describe a plan to determine the concentration of **FB 1**.

Your plan should include details of:

- calculation of the absorbance value for each copper(II) sulfate solution prepared in (b), given ϵ of CuSO₄ is 2.81 mol⁻¹ dm³ cm⁻¹.
- a sketch of the calibration line you would expect to obtain;
- a brief outline of how the results would be obtained.
- how the calibration line would be used to determine the concentration of copper(II) ions in **FB 1**.

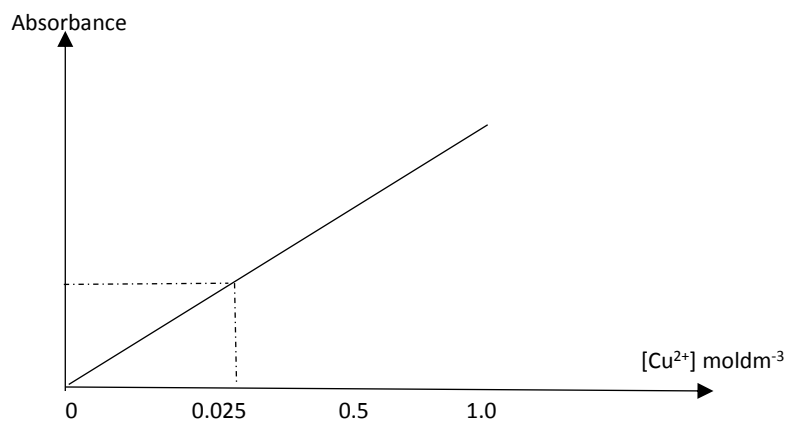
Table of results

Solution	[CuSO ₄] / mol dm ⁻³	Absorbance value
1	1.000	2.81
2	0.500	1.41
3	0.250	0.70
4	0.125	0.35
5	0.075	0.21
6	0.025	0.07

Using the spectrometer

1. Spectrometer is set up to absorb the wavelength of orange (600nm).

2. A few cm^3 of each of the 6 CuSO_4 solutions are run by the spectrometer and the absorbance values are recorded in Table 2.
3. Plot a graph of absorbance value versus $[\text{CuSO}_4]$.
4. Draw the best-fit straight line passing through the origin. This is the calibration line.



Analysing solution FB1

1. Run a few cm^3 of **FB1** in the spectrometer and record the absorbance value, A_x .
2. Using the graph drawn earlier, a horizontal line is drawn from this value to intersect the calibration line. By drawing a vertical line from the intersection point, $[\text{Cu}^{2+}]$ in **FB1** can be determined.

[3]

(d) Given that the absorbance for **FB 1** is 0.286, calculate the concentration of **FB 1**.

Applying Beer-Lambert's Law, $A = \epsilon cl$

$$0.286 = (2.81)(c)(1)$$

$$c = \underline{0.105 \text{ mol dm}^{-3}}$$

concentration of **FB 1** =[1]

[Total : 10]

Qualitative Analysis Notes

[ppt. = precipitate]

(a) Reactions of aqueous cations

cation	reaction with	
	NaOH(aq)	NH ₃ (aq)
aluminium, Al ³⁺ (aq)	white ppt. soluble in excess	white ppt. insoluble in excess
ammonium, NH ₄ ⁺ (aq)	ammonia produced on heating	–
barium, Ba ²⁺ (aq)	no ppt. (if reagents are pure)	no ppt.
calcium, Ca ²⁺ (aq)	white ppt. with high [Ca ²⁺ (aq)]	no ppt.
chromium(III), Cr ³⁺ (aq)	grey-green ppt. soluble in excess giving dark green solution	grey-green ppt. insoluble in excess
copper(II), Cu ²⁺ (aq),	pale blue ppt. insoluble in excess	blue ppt. soluble in excess giving dark blue solution
iron(II), Fe ²⁺ (aq)	green ppt., turning brown on contact with air insoluble in excess	green ppt., turning brown on contact with air insoluble in excess
iron(III), Fe ³⁺ (aq)	red-brown ppt. insoluble in excess	red-brown ppt. insoluble in excess
magnesium, Mg ²⁺ (aq)	white ppt. insoluble in excess	white ppt. insoluble in excess
manganese(II), Mn ²⁺ (aq)	off-white ppt., rapidly turning brown on contact with air insoluble in excess	off-white ppt., rapidly turning brown on contact with air insoluble in excess
zinc, Zn ²⁺ (aq)	white ppt. soluble in excess	white ppt. soluble in excess

(b) Reactions of anions

<i>ion</i>	<i>reaction</i>
carbonate, CO_3^{2-}	CO_2 liberated by dilute acids
chloride, $\text{Cl}^-(\text{aq})$	gives white ppt. with $\text{Ag}^+(\text{aq})$ (soluble in $\text{NH}_3(\text{aq})$)
bromide, $\text{Br}^-(\text{aq})$	gives pale cream ppt. with $\text{Ag}^+(\text{aq})$ (partially soluble in $\text{NH}_3(\text{aq})$)
iodide, $\text{I}^-(\text{aq})$	gives yellow ppt. with $\text{Ag}^+(\text{aq})$ (insoluble in $\text{NH}_3(\text{aq})$)
nitrate, $\text{NO}_3^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil
nitrite, $\text{NO}_2^-(\text{aq})$	NH_3 liberated on heating with $\text{OH}^-(\text{aq})$ and Al foil; NO liberated by dilute acids (colourless $\text{NO} \rightarrow$ (pale) brown NO_2 in air)
sulfate, $\text{SO}_4^{2-}(\text{aq})$	gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (insoluble in excess dilute strong acids)
sulfite, $\text{SO}_3^{2-}(\text{aq})$	SO_2 liberated with dilute acids; gives white ppt. with $\text{Ba}^{2+}(\text{aq})$ (soluble in dilute strong acids)

(c) Tests for gases

<i>gas</i>	<i>test and test result</i>
ammonia, NH_3	turns damp red litmus paper blue
carbon dioxide, CO_2	gives a white ppt. with limewater (ppt. dissolves with excess CO_2)
chlorine, Cl_2	bleaches damp litmus paper
hydrogen, H_2	"pops" with a lighted splint
oxygen, O_2	relights a glowing splint
sulfur dioxide, SO_2	turns aqueous acidified potassium manganate(VII) from purple to colourless

(d) Colour of halogens

<i>halogen</i>	<i>colour of element</i>	<i>colour in aqueous solution</i>	<i>colour in hexane</i>
chlorine, Cl_2	greenish yellow gas	pale yellow	pale yellow
bromine, Br_2	reddish brown gas / liquid	orange	orange-red
iodine, I_2	black solid / purple gas	brown	purple